Nischal A

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RESEARCH INTERESTS

Artificial Intelligence, Machine and Deep Learning, Natural Language Processing, Multi-modal Multimedia Analysis, Explainability and Robustness

EDUCATION BACKGROUND

Indian Institute of Technology Patna — **July.2018-May.2022**
Bachelor of Technology (B.Tech), Computer Science and Engineering — *CPI: 9.21/10*

VVS Sardar Patel PU College Bangalore — **May.2016-May.2018**
Higher Secondary in Physics, Chemistry, Maths, Electronics — *Percentage: 96.8%*

PUBLICATIONS

- **Nischal Ashok***, Rina Kumari*, Tirthankar Ghoshal and Asif Ekbal, **"A Multi-task Learning Approach for Fake News Detection: Novelty, Emotion, and Sentiment Lend a Helping Hand"**, International Joint Conference on Neural Networks, **IJCNN-2021**, (CORE-A) **Accepted** (* denotes joint-first authors)
- Rina Kumari*, **Nischal Ashok***, Tirthankar Ghoshal and Asif Ekbal, **"Misinformation detection using multitask learning with mutual learning for novelty detection and emotion recognition"**, Journal of Information Processing and Management, **IPM**, (IF:4.787) **Under Review** (* denotes joint-first authors)
- Rina Kumari*, **Nischal Ashok***, Tirthankar Ghoshal and Asif Ekbal, **"What the fake? Probing Misinformation Detection Standing on the Shoulder of Novelty and Emotion"**, Journal of Information Processing and Management, **IPM**, (IF:4.787) **Under Review** (* denotes joint-first authors)

EXPERIENCE

Undergraduate Researcher, Data and Knowledge Engineering Group — **Universität Wuppertal, Germany**
Guide: • Dr. Terry Ruas • Prof. Dr. Bela Gipp — *Dec.2020-Present*

- **Project:** Covid-19 Misinformation Detection
- "Pre-Trained" transformer based models on large corpus of Covid-19 data, CORD-19.
- Experimented with and state-of-the-art transformers based models on several Covid-19 datasets
- Work under review in a prestigious conference.

Undergraduate Researcher, AI-NLP-ML Lab — **IIT-Patna, India**
Guide: • Dr. Asif Ekbal • Prof. Pushpak Bhattacharya — *July.2019-Present*

- **Present Project:** Multi-modal (Video-Image-Text) based Misinformation Detection
- Procuring and analyzing video based data for fake news detection.
- Probed uni-modal misinformation detection through novelty detection, emotion recognition and sentiment analysis.
- Proposed a novel multi-tasking deep neural network architecture for fake news detection giving SOTA results on 4 well known datasets.
- **Previous Work:** Document Level Novelty Detection and IIT-Patna Academic Chatbot
- Co-authored papers in prestigious conferences and journals.

AI Research Intern, Video Analytics Laboratory — **Indian Institue of Science, India**
Guide: • Prof. Venkatesh Babu — *May.2020-Aug.2020*

- **Project:** Unlabelled Data improves Adversarial Robustness
- Investigated the role of unlabelled data for improving Adversarial Robustness in Deep Neural Networks using Cifar-10, Cifar-100, and TI-80M as the datasets.
- Experimented with the state-of-the-art adversarial deep learning algorithms based on semi-supervised learning.
- Proposed a novel method of generating unlabeled data from DC-GANs and DE-GANs to improve adversarial robustness. Journal Form reported accepted in the Indian Academy of Sciences Summer Research Portal.

Startup Winter AI Intern — **TildeHat, Bangalore, India**
Manager: • Dr. Om Deshmukh — *Dec.2019-Jan.2021*

- **Project:** Unsupervised Data Retrieval
- Designed a complete end-to-end module for useful information extraction from resumes.
- Implemented unsupervised algorithms and hierarchical classifier approaches for data-retrieval, conducting experiments on thousands of documents.

PROJECTS

IIT-Patna Academic Alexa — **IIT Patna**
Guide: Prof. Pushpak Bhattacharya, Former Director IIT-P — *Jan.2020-May.2020*

- A sentiment-aware intelligent information retrieval based Chatbot for the Academic Portal of IIT-Patna.

- Devised a novel and effective way of converting natural language data into SQL queries using Dependency Tree Parsers, POS tagging, and NER techniques
- Designed a custom chatbot pipeline which includes Multi-Channel CNN based intent classifier, sentiment analyzer, sentence similarity and feedback modules
- Designed a simple portable interface using wx-python GUI programming.

Reddit-Drug Abuse Detection **Data Analytics Lab, IIT Patna**
Guide: Dr. Joydeep Chandra *Jan.2020-May.2020*

- Investigated Drug Abuse on Social Media through Reddit Website by collecting and annotating data from r-opiates sub-reddit.
- Built a multi-head attention deep neural network for classifying comments as motivational or not. Further classification done on altruism, hope, good advice, bad advice, and universality.

Computer Science Academic Projects **IIT-Patna, India**
Guides: • Dr. Jimson Mathew • Dr. Samrat Mondal *May.2020-May.2021*

- Two Pass Assembler and Emulator - Computer Architecture CS321-322.
- Smart Lock System - Switching Theory CS225-226
- Delivery Company Database - Database Laboratory Course CS355
- HoneyPot based Authentication System - Secure System Design CS392

POSITIONS OF RESPONSIBILITIES

Head-Machine Learning/ AI **IIT Patna, India**
NJACK, Computer Science Club *May.2020-Present*

- Introduced NJACK ML Workshop Series comprising sessions of Competitive ML and Research in ML
- Taught students about topics like Ensemble ML, Introduction to Deep Learning and others.

Departmental Lead **IIT Patna, India**
DSC IIT Patna, Google Developers *Aug.2019-May.2020*

- Associate Facilitator of Explore-ML Workshop IIT-P powered by Google-AI
- Promoted and participated in projects like Drone-Eye. Introduced ML to freshmen students.

Head-Industrial Relations **IIT Patna, India**
Entrepreneurship Cell *Aug.2019-May.2021*

- Lead the IR division managing a team of 40 volunteers and 20 sub-coordinators.
- Introduced Entrepreneur-101 series and organized more than 4 sessions on entrepreneurship.

AWARDS and HONORS

- Summer Research Fellow (FASTSF'2020) - Indian Academy of Sciences [India's foremost Science Academy] for pursuing research internship in Artificial Intelligence at IISc, Bangalore.
- IIT-Patna Merit List - Among top 10 CPI holder in the IIT-Patna Computer Science batch 2018.
- Fellowship (KVPY'2016) - Kishore Vigyanik Protsahan Yojna - Government of India, for pursuing a career in research in pure sciences. Stood among top 0.2% among 200,000 students.
- Prathiba Ratna Award - Karnataka State (an Indian state) Education Development Authority for standing 2nd for the state in the 10th standard ICSE Board Exam 2016 among 150,000 students.

SKILLS

- **Programming Languages:** • C/C++ • Python • Java • Bash • Verilog • Assembly
- **Deep Learning Tools:** • Tensorflow • Keras • PyTorch • HuggingFace
- **Other tools:** • SQL • Docker • LaTeX • Git/GitHub
- **Operating Systems:** • Linux • Windows

SERVICE

- Institute Student Mentorship Program (2020-Present) - Academic guide to four freshmen year students at IIT-P over the period of two years.
- Social Worker. National Service Scheme - I served in the rural areas of the Bihar province of India, as a part of the rural entrepreneurship and awareness committee of the national service scheme
- Teaching Machine Learning to students of IIT-P. Link.
- Writing blogs on ML and AI. Link

CONFERENCES and TALKS

- Attendee - 43rd European Conference on Information Retrieval.
- Attendee - 17th International Conference on Natural Langauge Processing (ICON 2020)
- Attendee - AI for Social Impact Series - Pennsylvania State University